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Absolute zero is a temperature that an object can get arbitrarily close to but absolute zero will remain unattainable.

Thus the third law of thermodynamics states that the entropy of pure crystalline substance at absolute zero is zero. Otherwise it can be stated as it is impossible to lower the temperature of an object to absolute zero in a finite number of steps. Mathematically,

$\lim_{T \to 0} S = 0$ for a perfectly ordered crystalline state.

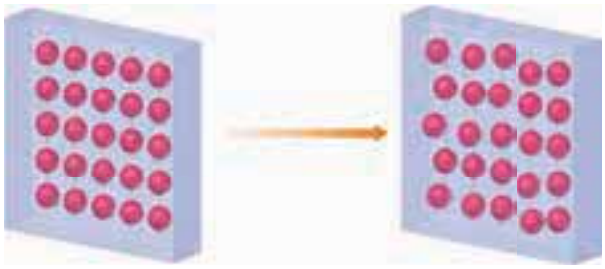


Figure 7.10 Third law of Thermodynamics

Crystals with defects (imperfection) at absolute zero, have entropy greater than zero. Absolute entropy of a substance can never be negative.